

5. An investigation was carried out into the productivity of grassland. Ecologists followed the method below.
- Three identical 1 m² areas of ungrazed grassland were selected.
 - All of the vegetation was removed from Area 1; the roots were washed and the vegetation dried.
 - Area 2 was covered with black plastic.
 - Area 3 was left undisturbed.
 - After two weeks, the vegetation was removed from Areas 2 and 3 and treated in the same way as Area 1.

The results were as follows:

Area	Dry mass / g m ⁻²
1	73
2	61
3	107

- (a) Explain why the dry mass was lower in Area 2. [2]

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- (b) Using the information from Areas 1 and 3, calculate the net primary productivity of the grassland.

Give your answer to 2 decimal places. [2]

Net primary productivity g m⁻² day⁻¹



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(c) Suggest why this method cannot be used to calculate the net primary productivity of tropical rainforest. [2]

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(d) To find the dry mass the ecologists dried the grass samples in an oven at 65 °C. Describe how the ecologists would be confident that all the water had been removed but none of the organic matter had been lost. [2]

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